

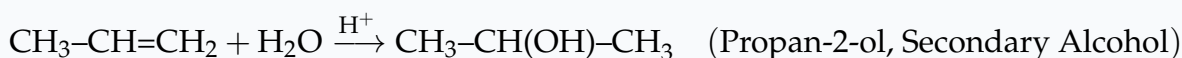
ALCOHOLS, PHENOLS AND ETHERS

Class 12th Chemistry · Core Name Reactions, Mechanisms & Chemical Tests

1. Preparation of Alcohols

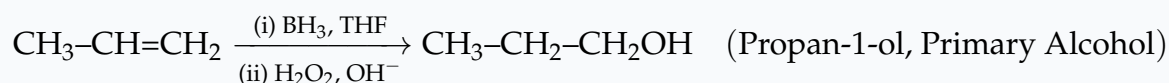
A. Acid-Catalyzed Hydration of Alkenes

Principle: Direct addition of water to alkenes in the presence of dilute mineral acid (H_2SO_4) proceeds according to **Markovnikov's Rule**. The electrophile (H^+) attacks to form a stable carbocation, followed by attack of nucleophilic water and deprotonation.



B. Hydroboration-Oxidation of Alkenes

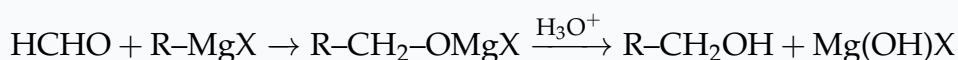
Principle: Indirect hydration of alkenes. The alkene reacts with diborane (B_2H_6 or $\text{BH}_3 \cdot \text{THF}$) to form a trialkylborane, which is subsequently oxidized by alkaline hydrogen peroxide ($\text{H}_2\text{O}_2/\text{OH}^-$). The net result is the addition of water across the double bond according to **Anti-Markovnikov's Rule**.



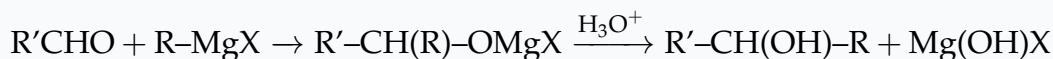
C. Synthesis of Alcohols from Grignard Reagent (R-MgX)

Principle: Nucleophilic addition of a Grignard reagent ($\text{R}^\delta - \text{MgX}^{\delta+}$) to the electrophilic carbonyl carbon ($\text{C}^\delta + = \text{O}^\delta -$) followed by acidic hydrolysis gives alcohols:

1. **From Formaldehyde (HCHO):** Produces a **Primary (1°) Alcohol**:



2. **From Any Other Aldehyde ($\text{R}'\text{CHO}$):** Produces a **Secondary (2°) Alcohol**:



3. **From Ketones ($\text{R}'_2\text{C=O}$):** Produces a **Tertiary (3°) Alcohol**:



2. Chemical Tests & Reactions of Alcohols

A. Lucas Test (Distinction between 1°, 2°, and 3° Alcohols)

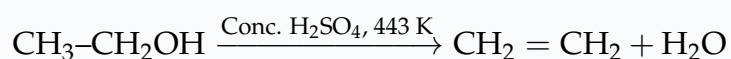
Reagent: Equimolar mixture of concentrated HCl and anhydrous ZnCl₂ (Lucas Reagent). It converts soluble alcohols into insoluble alkyl chlorides, producing turbidity.

- **Reactivity Order:** 3° > 2° > 1° (governed by carbocation stability).
- **3° Alcohol:** Reacts immediately; **turbidity appears instantly** at room temperature.
- **2° Alcohol:** Reacts moderately; **turbidity appears within 5 minutes**.
- **1° Alcohol:** Does not react at room temperature; **turbidity appears only upon heating**.

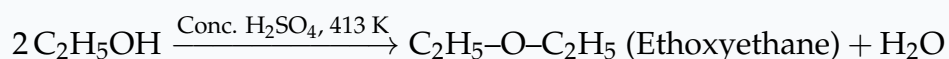
B. Acid-Catalyzed Dehydration of Alcohols

Heating alcohols with concentrated H₂SO₄ acts via an intramolecular or intermolecular pathway depending on temperature:

- **Formation of Alkene (Intramolecular Dehydration at 443 K):**

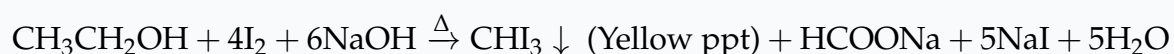


- **Formation of Ether (Intermolecular Dehydration at 413 K):**



C. Iodoform Test

Principle: Alcohols possessing a CH₃–CH(OH)– group (such as ethanol and all secondary methyl carbinols) react with iodine (I₂) in the presence of sodium hydroxide (NaOH) upon gentle heating to form a characteristic **yellow precipitate of iodoform (CHI₃)** alongside a carboxylate salt.

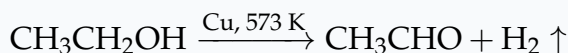


- Propan-2-ol gives a positive test; Propan-1-ol and tertiary alcohols give a negative test.

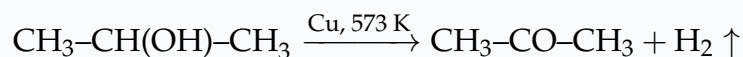
D. Catalytic Dehydrogenation over Hot Copper (Cu/573 K)

Passing alcohol vapors over reduced copper at 573 K provides a clean chemical distinction:

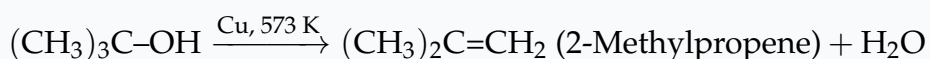
1. **1° Alcohol:** Undergoes **dehydrogenation** to form an **aldehyde**:



2. **2° Alcohol:** Undergoes **dehydrogenation** to form a **ketone**:



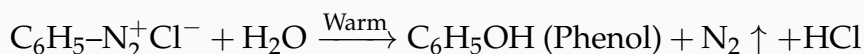
3. **3° Alcohol:** Lacks an α -hydrogen; undergoes **dehydration** instead of dehydrogenation to form an **alkene**:



3. Preparation of Phenol

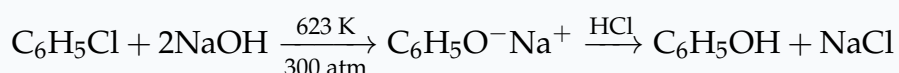
A. From Benzene Diazonium Chloride

Warming an aqueous solution of benzene diazonium chloride or passing steam through it hydrolyzes the diazonium group into phenol:



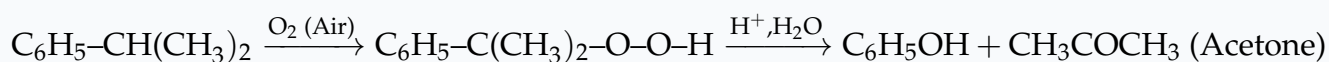
B. Dow's Process

Chlorobenzene is heated with aqueous NaOH at high temperature (623 K) and high pressure (300 atm) to yield sodium phenoxide, which upon acidification with HCl yields phenol:



C. From Cumene (Isopropylbenzene) — Industrial Method

Cumene is oxidized by air into cumene hydroperoxide, which on treatment with dilute acid yields phenol and an important commercial by-product, acetone:

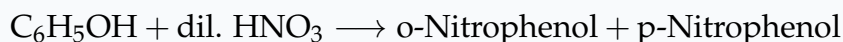


4. Electrophilic Substitution Reactions of Phenol

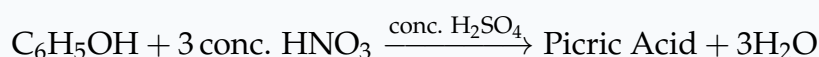
The --OH group is strongly activating and ortho/para-directing due to $+R$ resonance.

A. Nitration of Phenol

- **With Dilute HNO_3 at 298 K:** Forms a mixture of ortho- and para-nitrophenol, separable by steam distillation (ortho-nitrophenol is steam volatile due to intramolecular hydrogen bonding, while para-nitrophenol is less volatile due to intermolecular hydrogen bonding):



- **With Concentrated HNO_3 / H_2SO_4 :** Undergoes exhaustive substitution to yield **2,4,6-trinitrophenol (Picric Acid)**:



B. Bromination of Phenol

- **In Non-Polar Solvent (CS_2 or CHCl_3 at low temperature, 273 K):** Monobromination occurs yielding ortho- and para-bromophenol (para is major):

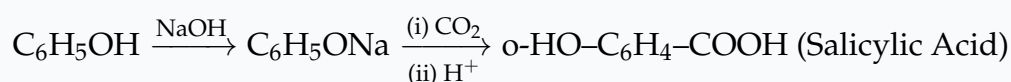


- **In Aqueous Medium (Bromine Water Test):** Phenol ionizes into strongly activated phenoxide ion, yielding an immediate **white precipitate of 2,4,6-tribromophenol**:



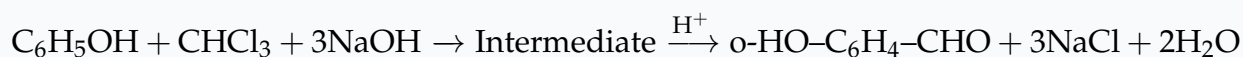
C. Kolbe's Reaction (Synthesis of Salicylic Acid)

Phenol is treated with sodium hydroxide to form phenoxide ion (which is more reactive than phenol toward electrophilic substitution), followed by reaction with carbon dioxide (CO_2 , a weak electrophile) under pressure (4 – 7 atm) at 400 K and subsequent acidification to yield **salicylic acid (2-hydroxybenzoic acid)**:



D. Reimer-Tiemann Reaction (Synthesis of Salicylaldehyde)

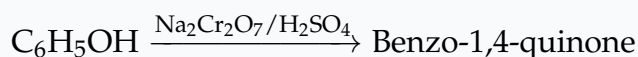
Phenol is treated with chloroform (CHCl_3) in the presence of aqueous sodium hydroxide, followed by acid hydrolysis, introducing an aldehyde ($-\text{CHO}$) group at the ortho position to form **salicylaldehyde (2-hydroxybenzaldehyde)**:



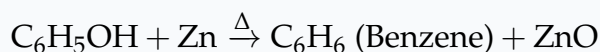
- **Electrophile:** Dichlorocarbene ($:\text{CCl}_2$), generated *in situ* from chloroform and base.

E. Oxidation and Reduction of Phenol

- **Oxidation with Chromic Acid ($\text{Na}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$):** Produces a conjugated diketone known as **benzoquinone**:



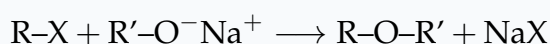
- **Reduction with Zinc Dust (Zn dust, Δ):** Phenol is reduced to **benzene**:



5. Preparation and Reactions of Ethers

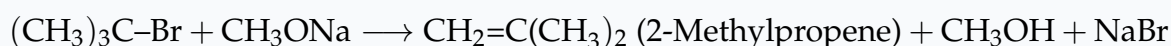
A. Williamson Ether Synthesis

Principle: An alkyl halide reacts with a sodium alkoxide/phenoxide to form symmetrical or unsymmetrical ethers via an $\text{S}_\text{N}2$ **mechanism**:



Crucial Limitation:

- The alkyl halide must be **primary** (1°).
- If a secondary (2°) or tertiary (3°) alkyl halide is used, elimination dominates over substitution due to the strong basicity of alkoxides, yielding an **alkene** instead of an ether:



B. Electrophilic Substitution of Anisole (Methoxybenzene)

The $-\text{OCH}_3$ group on anisole is ortho/para-directing:

- **Bromination:** $\text{Anisole} + \text{Br}_2 \xrightarrow{\text{CH}_3\text{COOH}} \text{4-Bromoanisole (Major)} + \text{2-Bromoanisole (Minor)}$
- **Nitration:** $\text{Anisole} + \text{HNO}_3/\text{H}_2\text{SO}_4 \longrightarrow \text{4-Nitroanisole (Major)} + \text{2-Nitroanisole (Minor)}$
- **Friedel-Crafts Alkylation:** $\text{Anisole} + \text{CH}_3\text{Cl} \xrightarrow{\text{Anhyd. AlCl}_3} \text{4-Methoxytoluene (Major)} + \text{2-Methoxytoluene}$
- **Friedel-Crafts Acylation:** $\text{Anisole} + \text{CH}_3\text{COCl} \xrightarrow{\text{Anhyd. AlCl}_3} \text{4-Methoxyacetophenone (Major)}$